

Your First AI-Integrated Python Project

A follow-along guide — from an empty API key to a working AI tool you built yourself

Before You Start

This assumes you already have Python 3 and VS Code set up (you do — it's in your `python-project` folder on the Desktop) and that you're comfortable with basic Python: variables, functions, loops. You don't need to know anything about AI or APIs yet — that's what this guide is for.

Step 1 — Get a Claude API Key

1. Create an account at `platform.claude.com`
2. Go to Settings → API Keys and generate a new key

This is a paid API (billed per use), but it's cheap — a small testing project like this one typically costs pennies. New accounts often come with some free trial credit. Treat this key like a password: never share it, never put it directly in your code, and never commit it to GitHub.

Step 2 — Set Up Your Project

Open a terminal in a new project folder and run:

```
mkdir ai-tool && cd ai-tool
python3 -m venv .venv
source .venv/bin/activate

pip install anthropic
```

A virtual environment (`.venv`) keeps this project's packages separate from everything else on your machine — good habit to start now.

Step 3 — Store Your Key Safely

Set your key as an environment variable instead of writing it into your code:

```
export ANTHROPIC_API_KEY="your-key-here"
```

The SDK automatically reads this variable — you never type your key into a `.py` file. If you turn this into a git repo later, add a `.gitignore` before your first commit so no secrets ever get tracked.

Step 4 — Your First API Call

Create `quickstart.py`:

```
import anthropic

client = anthropic.Anthropic()

message = client.messages.create(
    model="claude-opus-5",
    max_tokens=1000,
    messages=[
        {"role": "user", "content": "Explain recursion in one paragraph."}
    ],
)

for block in message.content:
    if block.type == "text":
        print(block.text)
```

Run it:

```
python quickstart.py
```

If you see a real response printed back — that's it. You just called an AI model from your own code. Everything past this point is building something useful on top of that same core pattern.

Step 5 — Turn It Into a Real Tool

A script that sends one hardcoded question isn't a project yet. Pick ONE of these and build toward it — don't try to do both:

Option A: Code Review Assistant

Reads a Python file you point it at and prints back structured feedback (bugs, style issues, suggestions).

```
import sys
import anthropic

client = anthropic.Anthropic()

filename = sys.argv[1]
with open(filename) as f:
    code_text = f.read()

message = client.messages.create(
    model="claude-opus-5",
    max_tokens=1000,
    messages=[{
        "role": "user",
        "content": f"Review this Python code and list concrete issues:\n\n{code_text}"
    }]
)

for block in message.content:
    if block.type == "text":
        print(block.text)
```

Run with: `python review.py main.py`

Option B: Study Buddy

Reads a text file of notes and generates a short quiz from it. Same shape as Option A — read a file, build a prompt around its contents, print the response — just a different prompt and purpose.

Either way, once the core works, level it up with:

- Command-line arguments instead of hardcoded filenames (the `sys.argv` pattern above)
- A loop so you can ask follow-up questions without restarting the program
- Error handling for a missing file or an API failure — same `try/except` pattern from your guessing game

What Comes After This

Once this works and you understand every line of it — not before — the natural next steps, roughly in order:

- **Give it a real interface:** a small Flask or FastAPI web app instead of a command-line script, with a database to save history, deployed live on something free like Render or Railway.
- **Push it to GitHub properly:** a real repo, a clear README explaining what it does and how to run it, meaningful commits.
- **Build an MCP server:** a small tool that lets an AI assistant interact with something real of yours — your notes, a database, an API. This is new enough that very few other junior candidates will have built one, which makes it a genuine differentiator rather than another AI-wrapper demo.

Troubleshooting

Error	What it means
AuthenticationError	Your API key isn't set correctly in this terminal session — re-run the export command.
ModuleNotFoundError : anthropic	Your virtual environment isn't activated, or the install failed — re-run <code>source .venv/bin/activate</code> then <code>pip install anthropic</code> .
RateLimitError	You're sending requests too fast for your account tier — add a short pause between calls.
Nothing prints, no error	Check you're actually looping over <code>message.content</code> and printing — a classic silent no-op, same shape as forgetting <code>()</code> on a method call.

Sources referenced for the setup steps: Anthropic's official API quickstart (platform.claude.com/docs/en/get-started).